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inventory = {}

while True:
    print("\n===== SMART INVENTORY MANAGEMENT SYSTEM =====")
    print("1. Add Product")
    print("2. Update Product")
    print("3. Search Product")
    print("4. Display Inventory")
    print("5. Low Stock Alert")
    print("6. Out of Stock Alert")
    print("7. Inventory Report")
    print("8. Delete Product")
    print("9. Exit")

    choice = int(input("Enter your choice: "))

    # Add Product
    if choice == 1:
        pid = input("Enter Product ID: ")
        name = input("Enter Product Name: ")
        category = input("Enter Category: ")
        qty = int(input("Enter Quantity: "))
        price = float(input("Enter Price: "))
        supplier = input("Enter Supplier: ")

        inventory[pid] = {
            "Name": name,
            "Category": category,
            "Supplier": supplier
        }

        print("Product Added Successfully!")

    # Update Product
    elif choice == 2:
        pid = input("Enter Product ID: ")

        if pid in inventory:
            inventory[pid]["Quantity"] = int(input("New Quantity: "))
            inventory[pid]["Price"] = float(input("New Price: "))
            print("Product Updated Successfully!")
        else:
            print("Product Not Found!")

    # Search Product
    elif choice == 3:
        pid = input("Enter Product ID: ")

        if pid in inventory:
            print(inventory[pid])
        else:
            print("Product Not Found!")

    # Display Inventory
    elif choice == 4:
        if len(inventory) == 0:

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        print("\nInventory Details:")
        for pid, details in inventory.items():
            print("\nProduct ID:", pid)
            for key, value in details.items():
                print(key, ":", value)

# Low Stock Alert
elif choice == 5:
    threshold = 10

    print("\nLow Stock Products:")
    found = False

    for pid, details in inventory.items():
        if details["Quantity"] < threshold:
            print(pid, "-", details["Name"])
            found = True

    if not found:
        print("No Low Stock Products")

# Out of Stock Alert
elif choice == 6:
    print("\nOut of Stock Products:")
    found = False

    for pid, details in inventory.items():
        if details["Quantity"] == 0:

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        if not found:
            print("No Out of Stock Products")

# Inventory Report
elif choice == 7:
    total_items = 0
    total_value = 0

    for details in inventory.values():
        total_items += details["Quantity"]
        total_value += details["Quantity"] * details["Price"]

    print("\n===== INVENTORY REPORT =====")
    print("Total Items:", total_items)
    print("Total Inventory Value:", total_value)

# Delete Product
elif choice == 8:
    pid = input("Enter Product ID to Delete: ")

    if pid in inventory:
        del inventory[pid]
        print("Product Deleted Successfully!")
    else:
        print("Product Not Found!")

# Exit
elif choice == 9:
    print("Thank You!")
    break

else:
    print("Invalid Choice!")

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Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
- RESTART: C:/Users/imtiy/AppData/Local/Programs/Python/Python311/noorejara_task9_(TENPYTHON DEVELOPE
R1159).py

===== SMART INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Update Product
3. Search Product
4. Display Inventory
5. Low Stock Alert
6. Out of Stock Alert
7. Inventory Report
8. Delete Product
9. Exit
Enter your choice: 1
Enter Product ID: 111
Enter Product Name: cream
Enter Category: beauty
Enter Quantity: 345
Enter Price: 50000
Enter Supplier: zara
Product Added Successfully!
```